

Maitreyee Sar

she/her

Bachelor of Technology in
Computer Science and Engineering
Kalinga Institute of Industrial Technology (KIIT)

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ACADEMIC DETAILS

Examination	Major	Year	Grade
Bachelor of Technology	<i>Computer Science and Engineering</i>	2022–Present	9.75/10 [Rank: 6/4276]
Higher Secondary (AISSCE, PCM)	—	2022	93.8%
Secondary (ICSE)	—	2020	96.4%

PUBLICATIONS

- **Peel and Pool: The Path to Mandala Perfection.** Tusita Sarkar, [Maitreyee Sar](#), Partha Bhowmick. *International Conference on Pattern Recognition (ICPR), December 2024 in Kolkata, India.* [Link]
- **Predicting Spatiotemporal Delays in Aviation Networks using Graph Neural Networks.** Maitreyee Sar, Chittaranjan Pradhan. *International Conference on Computer, Electrical & Communication Engineering, February 2026 in Kolkata, India* [Link]

POSTERS/PRESENTATIONS

- **Poster: Modeling Network Behaviour and Interference in HPC Networks at Scale.** Presented at IIT Kanpur, India July 2025
- **Predicting Spatiotemporal Delays in Aviation Networks using Graph Neural Networks.** Presented at the International Conference on Computer, Electrical & Communication Engineering, Kolkata February 2026

INTERNSHIPS

- **[Ongoing] Geometric Evaluation of 10-metre Foundation Model Embeddings for Crop Phenology in CONUS**
Guide: Dr. Adway Mitra, IIT Kharagpur Feb 2026 - Present
- **Modeling Network Behaviour and Interference in HPC Networks at Scale | C/C++, OpenMPI, SST** Guide: Dr. Preeti Malakar, IIT Kanpur May-July 2025
 - Analyzed the effect of different network topologies and communication patterns on performance in HPCs
 - Conducted thorough literature survey on various simulation frameworks for exascale, coordinated with original authors to resolve minor implementation bugs
- **Geometric Analysis of Mandala in the Context of Computational Art | Python, OpenCV** Guide: Prof. Partha Bhowmick, IIT Kharagpur May-July 2024
 - Implemented an algorithm to process handcrafted Mandala artwork using binarization and connected-component analysis with a “peeling” technique to detect its order of symmetry and correct geometric distortions

- Collected statistical data on performance variations for different levels of size, density and distortions in input images
- **Keyword-based Document Indexing and Retrieval System** | *Python, SQL* June-Aug 2023
Guide: Prof. Arobinda Gupta, IIT Kharagpur
 - Developed a file indexing and retrieval system with a MySQL-based caching mechanism to optimize repeated searches and reduce lookup time

ACADEMIC ACHIEVEMENTS

- Qualified **Graduate Aptitude Test in Engineering (GATE)** 2026 and 2025, in CS and DA
- **TOEFL** (November 2025): 115/120. **GRE** (November 2025): 163Q 156V
- Selected for **Summer Research Fellowship 2025** by the *Indian Academy of Sciences, Indian National Science Academy, and The National Academy of Sciences*, India
- Selected for **Summer Research Fellowships** at **IIT Kanpur (Institute-Funded category)**, **IIT Delhi**, and **IIT Madras**, India, 2025

PROJECTS

- **Predicting Spatiotemporal Delays in Aviation Networks using Graph Neural Networks** | *Python (PyTorch, PyTorch Geometric)* 7th Semester Project, November 2025 Developed a Graph Attention Neural Network with LSTM to capture the spatio-temporal dynamics of a network of airports.
- **mnemo: A RAG-Based Memory Aid** | *Python, SQLite, FAISS, HuggingFace, Ollama* 6th Semester Mini Project, April 2025 Developed a personal memory assistant for individuals with memory loss, using embeddings and FAISS for similarity search with a dedicated people database. Currently integrating auto-generated keywords and multimodal input (audio, images etc). **[GitHub]**
- **solver: Algorithm Solver and Visualizer** *Python (Flask, Matplotlib, NumPy, NetworkX), HTML, CSS, JavaScript* Applications Development Lab Project, April 2025 Built a Flask-based visualization and solver tool for common algorithmic paradigms (Greedy, Dynamic Programming, Graph). Currently expanding with real-time visualizations and advanced algorithm modules. **[GitHub]**

TECHNICAL SKILLS

- **Programming Languages.** Python, Java, C
- **Frameworks.** scikit-learn, PyTorch, OpenCV, Flask, Web development tools
- **Other.** SQL, OpenMPI, Git, Linux, Latex

COCURRICULAR ACTIVITY

- **Student Volunteer** at the **ACM Women in Computing Grad Cohort 2025**, IIT Kanpur, India
- Student Member, IEEE
- English Editor at Wordsmith, KIIT Literary Society, Founding Members' Team for Inkspire, the Wordsmith blog
- **Dramatics Minister**, Carmel Convent High School (2020 batch), Durgapur, India: Student-in-charge for conducting school events, anchoring, and scriptwriting
- **Sangeet Visharad Diploma** (Vocal Classical Music, 5 Years, First Division with Distinction), 2022 — Pracheen Kala Kendra, Chandigarh, India
- **Ankan Bivakar Diploma** (Painting, 5 years, First Division with Distinction), 2010-2022 — Bangiya Sangeet Parishad, affiliated with Rabindra Bharati University, Kolkata, India